

Sentencing Disparities between Sex and Race

Kristian A. Nunez

University of Central Florida

8 July 2025

Research Questions

- How do sentencing disparities by **sex** and **race** manifest in Florida homicide cases?
- Do disparities vary across **sentencing guideline eras** (Pre-1983, 1983, 1994, 1995, CPC)?
- Are observed disparities driven by:
 - Differences in case characteristics (*explained component*)?
 - Judicial bias or discretionary factors (*unexplained component*)?
- How do sentencing outcomes differ for:
 - Fixed-term sentences vs. life or death sentences?
 - Death sentence vs. life sentence among capital cases?
 - Sentence severity within fixed-term cases?

Sentencing Guidelines as Institutional Context

- Florida's sentencing system has evolved through multiple guideline eras:
 - **Pre-1983:** Unstructured sentencing, high judicial discretion.
 - **1983, 1994, 1995 Guidelines:** Increasing formalization and constraint.
 - **Criminal Punishment Code (CPC), 1998:** Current structured sentencing framework.
- Each era imposes distinct rules affecting judicial discretion and sentencing severity.
- Institutional shifts create a natural experiment to observe if and how disparities by sex and race change over time.
- We explicitly model interactions between sex, race, and guideline eras to test evolving disparities.

Data Cleaning Workflow — Key Steps

- **Merge datasets** using unique IDs to consolidate offender records
- **Fill missing demographics** (Race, Sex) from overlapping datasets
- **Calculate age** at offense and adjudication events, flag minors
- **Categorize sentencing systems** based on offense dates
- **Assign felony degrees** from charge categories

Data Cleanup and Finalization

- **Remove duplicates** based on key columns (DCNumber, CaseNumber, Sequence)
- **Drop unnecessary columns** to simplify datasets
- **Standardize column names** for consistency
- **Handle special sentencing cases**, e.g. life/death sentences
- **Export cleaned datasets** ready for analysis

Sentencing System Standardization - Overview

- Applies Florida's sentencing guidelines to cleaned homicide offender data from 3 datasets
- Computes standardized min/max sentence durations (in days) based on applicable sentencing system
- Sentencing systems included:
 - Unstructured Sentencing (pre-1983)
 - 1983, 1994, 1995 Florida Sentencing Guidelines
 - Criminal Punishment Code (CPC)
- Output files: cleaned datasets with sentence duration ranges

Sentencing System Standardization - Details

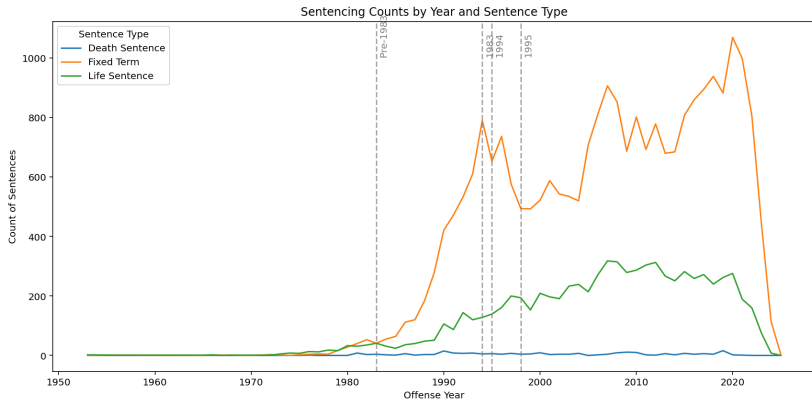
- Maps charges and felony degrees to offense levels and sentencing points
- Calculates total sentence points from:
 - Primary, secondary, and prior offenses
 - Legal status
 - Injury and weapon involvement
- Converts points into estimated prison time ranges (days)
- Enables consistent analysis across historical sentencing frameworks

Analysis Overview

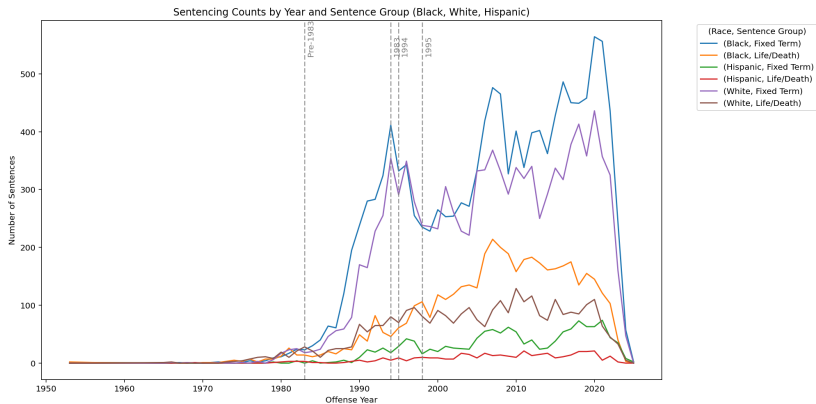
- Explores sentencing patterns in Florida homicide cases, focusing on life, death, and fixed term sentences.
- Evaluates sentencing severity variations across demographic groups and guideline eras.
- Tests for racial and sex disparities in sentencing.
- Applies selection correction (Heckman model), Oaxaca-Blinder decomposition, and RIF regressions for quantile disparities.
- Visualizations illustrate sentencing trends, disparities, and model results.

- Trends analysis over time
- Logistic regression model for life sentence outcomes
- OLS regression for sentence severity (fixed term sentences)
- Manual two-step Heckman selection correction to address selection bias
- Oaxaca-Blinder decomposition for analyzing racial sentencing disparities
- RIF regression for examining disparities across sentence quantiles

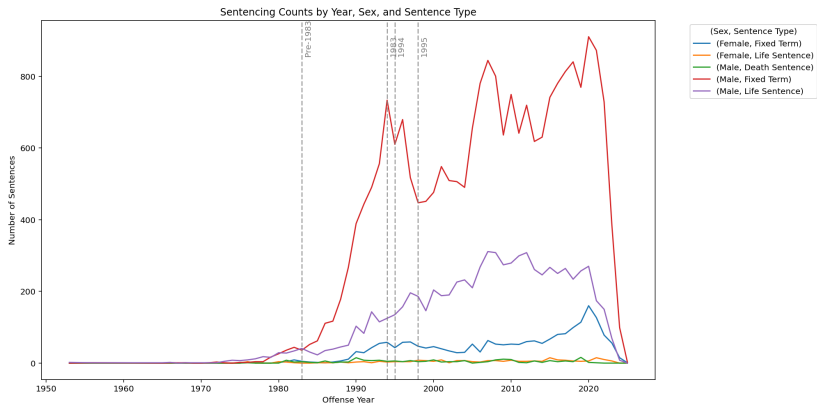
Trend 1



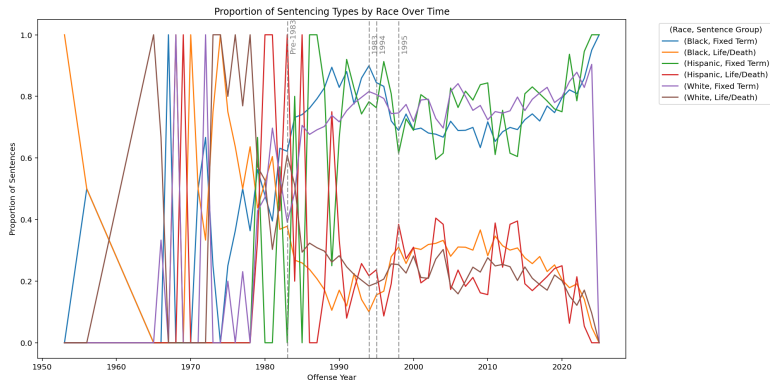
Trend 2



Trend 3

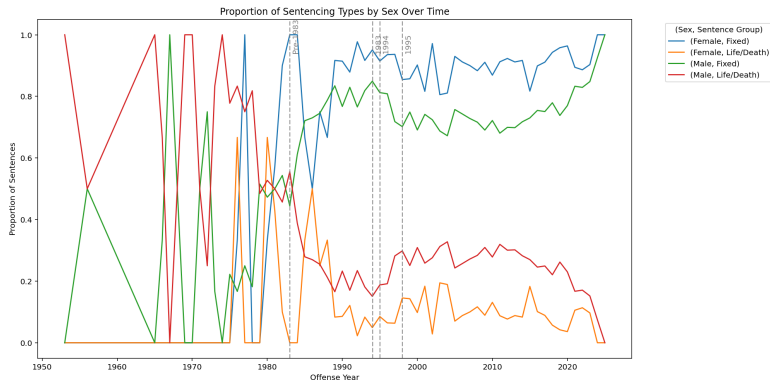


Trend 4: Sentencing by Race and Era



- Pre-1983: sentencing proportions varied widely by race.
- Post-1983 guidelines: more stable, consistent sentencing.
- Fixed sentences rose; Life/Death sentences declined.

Trend 5: Sentencing by Sex and Era

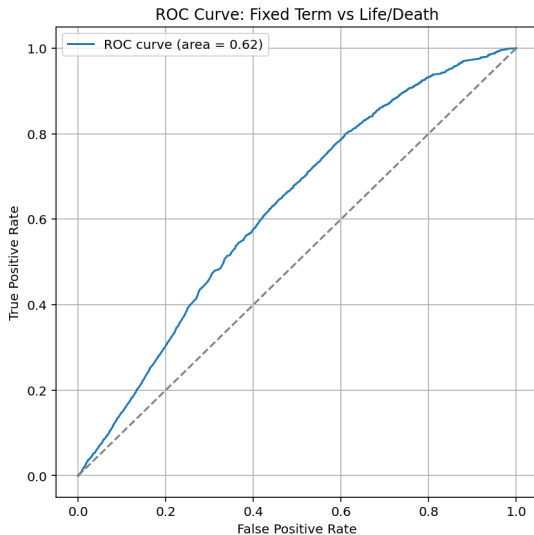


- Before 1983, females got more Fixed terms; males more Life/Death.
- Post-1983, proportions stabilized with less variance.
- Life/Death sentences still more common for males.

Logistic Model: Life vs Fixed/Death Sentences

- **Sample:** 36,266 cases; Dependent variable: `is_life`
- **Model fit:** Pseudo $R^2 = 0.034$ (significant but limited)
- **Key predictors:**
 - **Male** offenders more likely to receive life sentences.
 - **Black** race increases life sentence odds vs White.
 - Post-1983 guideline eras → higher life sentence odds.
 - **Older age** → higher odds; **More charges** → lower odds.
- **Note:** Model shows significant associations but low explanatory power.

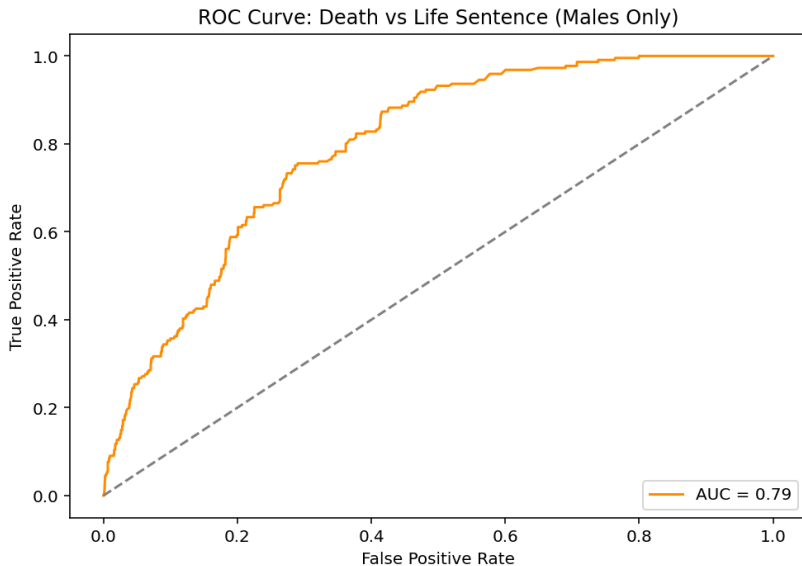
AUC of Fixed vs Life/Death



Logistic Model: Death vs Life (Males Only)

- **Sample:** 7,885 male-only cases; Dependent variable: `is_death`
- **Model fit:** Pseudo $R^2 = 0.119$ (moderate)
- **Key predictors:**
 - **Black race** → lower odds of death vs White.
 - **Later guideline eras** → lower death sentence odds.
 - **1st degree felony** → strongly reduces death sentence likelihood.
 - **Capital charge, age, charge count** → increase death sentence odds.
- **Note:** Only males included (no females on death row). Results are exploratory.

AUC of Life Vs Death (Males Only)



OLS Model: Sentence Severity (Fixed-Term Sentences Only)

Model Summary:

- **Dependent Variable:** Sentence severity (standardized)
- **Sample:** 36,266 fixed-term sentence cases
- $R^2 = 0.167$ — Model explains 16.7% of variance
- **Controls:** Race, Sex, Guideline Era, Charge Category, Age, Repeat Status, Charge Count

Key Insight: Sentencing severity is shaped most strongly by guideline era and offense type.

OLS Key Findings: Demographics, Charges, and Guidelines

Demographics:

- **Sex (Female):** Lower severity (↓)
- **Race:** No significant impact overall

Guideline Eras:

- All post-1983 eras $\Rightarrow$ lower severity vs. pre-1983
- Suggests greater consistency and less harshness over time

Charge Categories:

- **Attempted Murder, Manslaughter** $\Rightarrow$ Higher severity
- **DUI/Gun Homicides:** Less effect on sentence length

Other Factors:

- **Repeat Offenders:** Receive lighter sentences (possibly due to plea deals or charge stacking)
- **Age at Offense:** Older offenders sentenced more harshly

Sentence Severity: First-Time vs. Repeat Offenders

Key Observation:

- **First-time offenders** receive **harsher sentences** on average.
- Mean severity:
 - First-time: **-5.74**
 - Repeat: **-10.15**
- Difference is **statistically significant** ($t = 56.31, p < 0.001$)

Interpretation:

- First offenses often involve more serious crimes.
- Repeat offenders may face **less severe** terms due to:
 - Charge stacking or plea deals
 - Judicial discretion within guideline limits

Heckman Selection Model: Sentence Severity

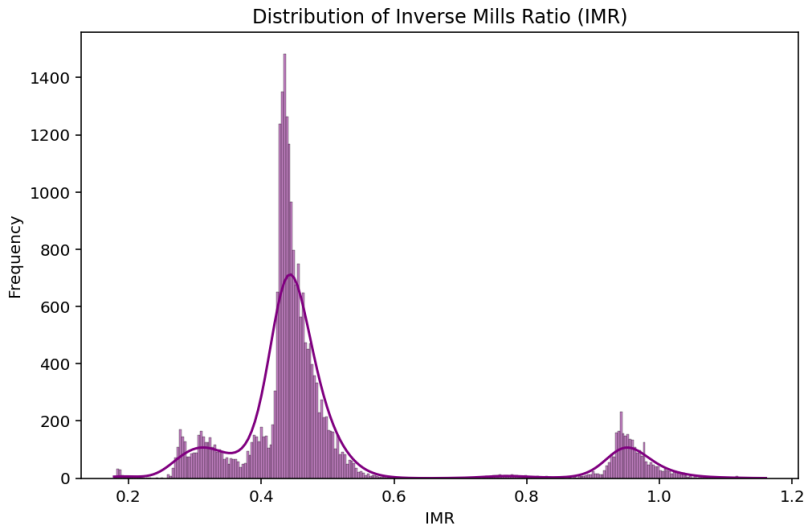
Goal: Not every offender has a clear sentence length. This model predicts *who gets a measurable sentence*.

Key Findings:

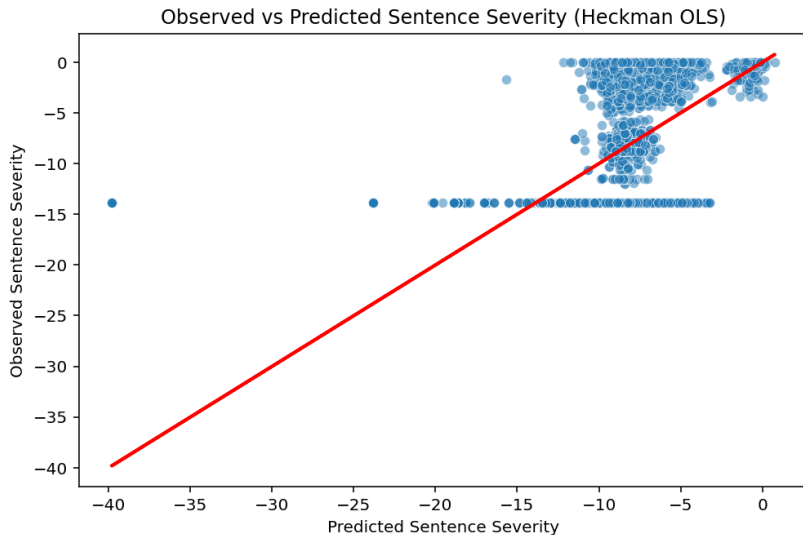
- **Younger offenders** are more likely to have sentence severity recorded.
- **More charges** → more likely to have a recorded sentence.
- **Men** are *less* likely to have sentence severity recorded.
- **Recent sentencing eras** (1994, 1995, CPC) → more likely to have measurable sentences.

Takeaway: There is selection bias — not everyone is included equally. The *Inverse Mills Ratio* confirms this bias exists.

Distribution of Inverse Mills Ratio



Observe vs Predicted Severity)



Heckman Outcome Equation: Sentence Severity

This step predicts sentence severity, only for offenders with known sentence length. **Key Drivers:**

- **Older age at offense** → longer sentences
- **More charges** → shorter sentences per charge (maybe due to plea deals)

Sentencing Era Effects:

- **Pre-1983** → harshest sentences
- **1994** → more severe than modern (CPC)
- **1995** → less severe than CPC

Race Gender:

- **Men** get slightly harsher sentences (not statistically strong)
- **Black offenders** get slightly lower sentences (weak evidence)
- **No strong racial disparities** after accounting for other factors

Note: These are average effects, not individual-level conclusions.

Racial Disparities: Oaxaca-Blinder Decomposition

Focus: Black vs. White defendants (Fixed-term sentences only)

Average Severity (log scale):

- Black: -8.19
- White: -7.77

Total gap: -0.41 (Black receive harsher sentences on average)

What explains the gap?

- **63%** is due to observed factors (age, charge count, sex, guideline era)
- **37%** is unexplained — could reflect unmeasured factors or systemic disparities

Takeaway: While most of the racial gap is explained by case differences, a meaningful portion remains unexplained.

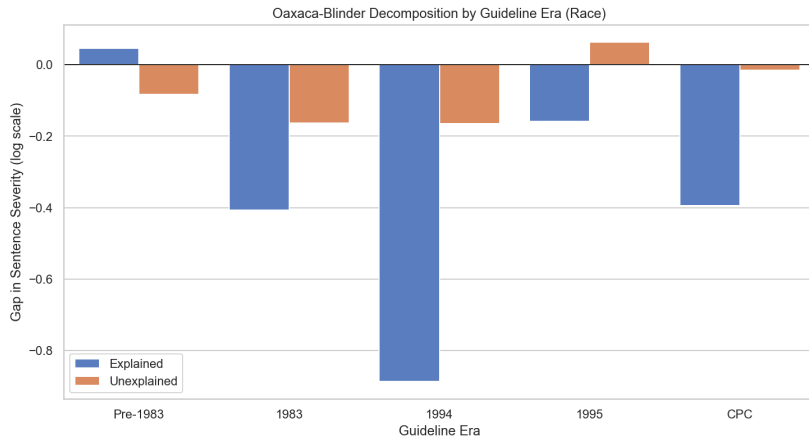
Racial Disparities Over Time (Black vs. White)

Oaxaca Decomposition by Guideline Era:

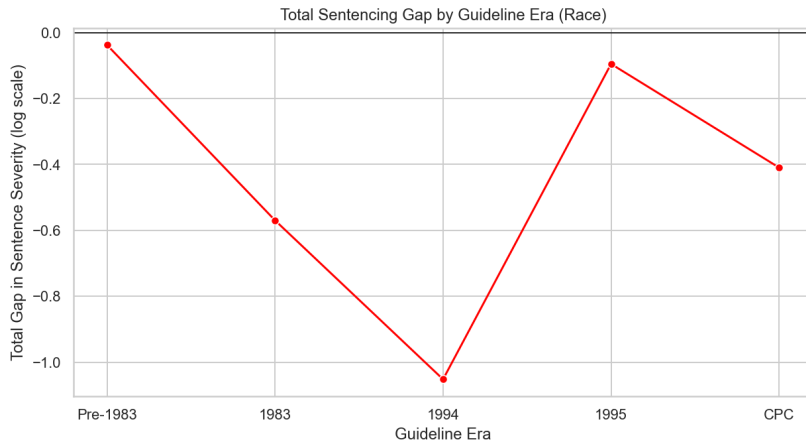
- **Pre-1983:** Small gap, small unexplained difference
- **1983:** Larger total gap, **29% unexplained**
- **1994:** Largest gap overall, **16% unexplained**
- **1995:** Very small gap, unexplained part favors Black defendants
- **CPC (Post-1998):** Gap mostly explained by case differences

Conclusion: Racial disparities were more pronounced during 1983 and 1994. The CPC era shows improved equity.

Oaxaca-Blinder Decomposition by Guideline (Race)



Total Sentencing Gap (Race)



Sex Disparities: Oaxaca-Blinder Decomposition

Focus: Male vs. Female defendants (Fixed-term sentences only)

Average Severity (log scale):

- Male: -7.98
- Female: -8.18

Total gap: +0.21 (Males get harsher sentences on average)

Breakdown:

- **Explained:** -0.96 — men “should” have gotten even lower severity
- **Unexplained:** +1.16 — women receive harsher sentences than expected

Takeaway: Women may face stricter sentencing after controlling for case differences.

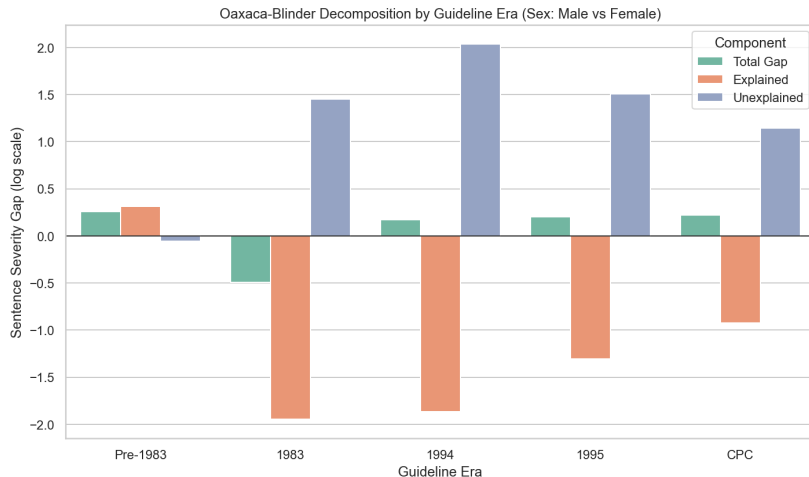
Sex Disparities Over Time (Male vs. Female)

Oaxaca Decomposition by Guideline Era:

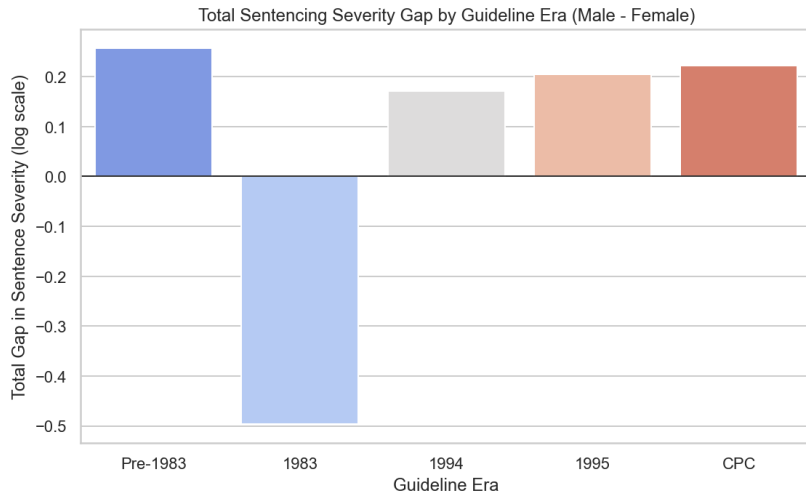
- **Pre-1983:** Small unexplained difference favoring females
- **1983, 1994, 1995, CPC:**
 - Males have higher severity on average
 - **Unexplained gap favors females in every era**
 - Women receive harsher-than-expected sentences after controlling for case details

Conclusion: Post-1983, sentencing gaps by sex increasingly favor men after adjusting for case characteristics.

Oaxaca-Blinder Decomposition by Guideline (Sex)



Total Sentencing Gap (Sex)



RIF Regression at the Median Sentence Severity

Goal: Examine how race, sex, age, and charge count affect sentence severity at the **50th percentile (median)**.

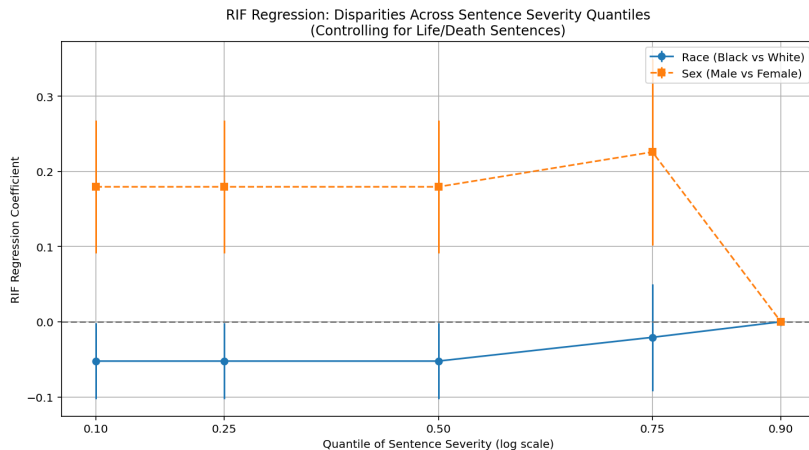
Key Results:

- **Race (Black vs. White):** Black defendants received slightly less severe sentences at the median (coef = -0.052, $p = 0.043$).
- **Sex (Male vs. Female):** Males received significantly more severe sentences (coef = +0.180, $p < 0.001$).
- **Age at offense:** Older offenders received harsher sentences (coef = +0.011, $p < 0.001$).
- **Charge count:** More charges were linked to *slightly shorter* sentences (coef = -0.049, $p < 0.001$).

Model Fit: $R^2 = 0.033$ (modest explanatory power)

Note: Life and death sentences excluded; analysis limited to fixed terms.

RIF Regression

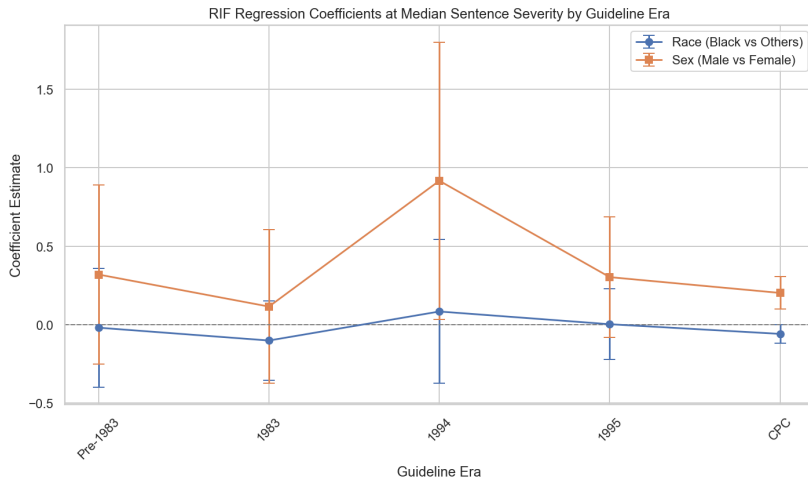


RIF Regression by Sentencing Era (Median Severity)

RIF Models run separately for each guideline era:

- **Pre-1983:** $R^2 = 0.078$
 - Race/sex effects not significant
 - Age $\uparrow$, charge count $\downarrow$ severity
- **1983:** $R^2 = 0.110$
 - No race/sex effect
 - Age $\uparrow$, charge count $\downarrow$
- **1994:** $R^2 = 0.043$
 - Males received harsher sentences
 - Age $\uparrow$, charge count $\downarrow$
- **1995:** $R^2 = 0.059$
 - No race/sex effect
 - Age $\uparrow$, charge count $\downarrow$
- **CPC Era:** $R^2 = 0.030$
 - Race marginally significant (Black $\rightarrow$ slightly lower severity)
 - Sex and age $\uparrow$, charge count $\downarrow$

RIF Regression Across Guideline Eras



RIF Regression: Key Takeaways

Across all eras:

- **Age:** Consistently linked to harsher sentences
- **Charge count:** Linked to *less* severity — possibly due to plea deals or charge stacking

Race Effects:

- Generally not significant
- Slight negative effect (shorter sentences) for Black defendants in CPC era

Sex Effects:

- Mixed results, but males received significantly harsher sentences in some eras (1994, CPC)

Model Fit:

- Explanatory power is modest across all eras (R^2 between 0.03 and 0.11)
- Suggests other factors — judicial discretion, plea deals, unobserved variables — may play a major role

Conclusion: Do Disparities Exist in Florida Homicide Sentencing?

Race:

- No consistent evidence of systemic bias in fixed-term sentence severity.
- Logistic models suggest Black defendants are *more* likely to receive life sentences.
- Oaxaca-Blinder: Racial gaps in sentence severity largely explained by observed characteristics, but some unexplained disparity persists — especially in 1983 and 1994 eras.
- RIF regressions: Race effects generally weak or non-significant, with small differences at the median.

Sex:

- Women receive shorter fixed-term sentences overall.
- However, Oaxaca-Blinder shows women face *unexplained* harsher sentencing after 1983 when controlling for covariates.
- Men are more likely to receive life or death sentences (logistic models).
- RIF: Male defendants receive significantly harsher median sentences in some eras (e.g., 1994, CPC).